

```

import numpy as np
import pandas as pd
from sklearn.ensemble import ExtraTreesRegressor
import matplotlib.pyplot as plt
import joblib

# load dataset
df = pd.read_csv("dataset.csv")
df["Label"] = df["Year"].astype(str) + "-" + df["Month"].astype(str).zfill(2)

# detect the latest recorded month and year
latest_year = df["Year"].max()
latest_month = df[df["Year"] == latest_year]["Month"].max()

# train using data up to latest month
train_df = df[(df["Year"] < latest_year) | ((df["Year"] == latest_year) & (df["Month"] <= latest_month))]
X_train = train_df[["Year", "Month", "Median_Temp", "Median_Humidity"]].values
y_train = train_df["Total_kWh"].values

# train model
model = ExtraTreesRegressor(n_estimators=200, max_depth=6, random_state=42)
model.fit(X_train, y_train)

# predict for the rest of the year
if latest_month < 12:
    months_remaining = list(range(latest_month + 1, 13))
    future_df = pd.DataFrame({
        "Year": [latest_year] * len(months_remaining),
        "Month": months_remaining
    })
    recent = df.tail(6)

```

Fig. 8: Snippet of machine learning model code

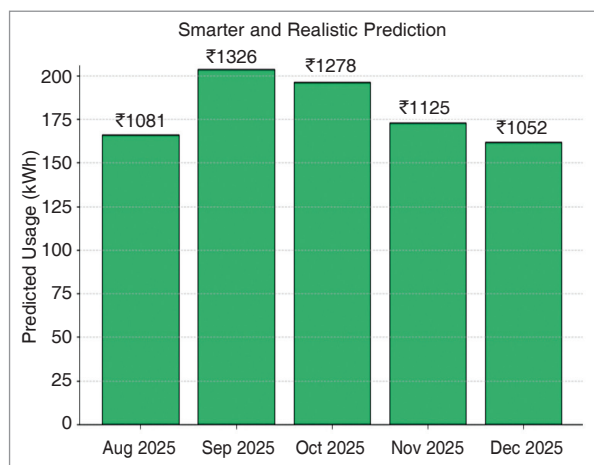


Fig. 9: Machine learning model prediction for the remainder of the year

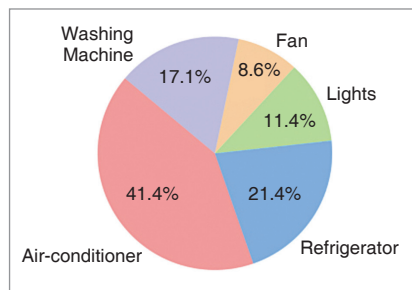


Fig. 10: Appliance-wise energy contribution

employs a machine learning model called the random forest regressor. This model is selected because it handles non-linear datasets effectively and has a lower risk of

overfitting, resulting in more accurate predictions for future data.

Initially, the model is trained on historical data and subsequently logs new data from the ESP32 at the end of each month. Fig. 8 shows a snippet of the ML model code, which was implemented using Pandas and NumPy in Jupyter Notebook.

Pandas and NumPy are fundamental Python libraries widely used for data science and numerical computing, while Jupyter Notebook provides an interactive environment that allows step-by-step code execution and instant visualisation of results, including graphs and predictions.

Fig. 9 presents the graph generated when the 'Remaining Year Prediction' button was clicked.

Appliance-level monitoring and scalability

The current system monitors

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energy usage at the household level. However, its functionality can be expanded to track individual appliance consumption. This can be achieved by deploying multiple sensor nodes at key outlets or by using Wi-Fi-enabled smart plugs.

A central user interface can then display appliance-specific usage data and generate a monthly report. This feature allows identification of energy-intensive devices and supports the adoption of energy-saving practices. Fig. 10 shows the appliance-wise energy contribution.

Taking control of your energy future

This system provides an accessible and effective method for gaining valuable insight into home energy consumption. By combining the versatility of the ESP32 with the predictive power of a machine learning model, it not only monitors electricity but also enables informed decision-making that can significantly influence monthly bills.

The energy monitor extends beyond basic tracking, offering a structured approach to achieving greater energy efficiency and cost-effectiveness. Building this system marks a significant step towards smarter energy management and meaningful savings. **EFY**

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